



AgriConnect – Offline-First Farmer Marketplace & Advisory Platform

Team ID: 8 | Section: S-3 | Members: M. Likhitha – 2520030575, K. Vaishnavi – 2520030198, S. Nikhil – 2520030246 | Guide: Dr. S. Spandana

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Abstract

AgriConnect is a web platform that helps farmers and sellers manage crops, expenses, inventory, market prices, and sales. It uses HTML, CSS, JavaScript, Node.js, Express.js, and SQLite. **Benefit:** Makes farm management easier and connects farmers with buyers and markets.

Problem Statement & Objectives

Problem: Farmers lack one simple platform to manage farming and selling.
Objective 1: Manage crops and farm activities.
Objective 2: Track expenses, profits, and market prices.
Objective 3: Connect farmers with buyers.
Scope: Farmers and sellers; crops, finance, marketplace, inventory, and orders.

Existing vs Proposed System

Feature	Existing System	Proposed System
Farm Management	Separate/manual	All-in-one platform
Expense Tracking	Manual records	Digital tracking
Market & selling	Limited access	Connect with buyers
Inventory & Orders	Not integrated	Integrated management
Limitation / Gap	No offline-first solution	Offline-first support

Feasibility Study

Technical: HTML, CSS, JavaScript, Node.js, SQLite
Economic: Low cost using open-source tools
Operational: Simple, user-friendly, and easy to maintain
Schedule: Completed within the semester timeline

Results & Testing

Outcomes: Efficient farm management.
CRUD: Add, view, edit, delete.
Validation: Input validation.
Performance: Fast response.

Conclusion

Achieved: Complete farm management platform.
Objectives: Simplifies farming, finance, and marketplace activities.

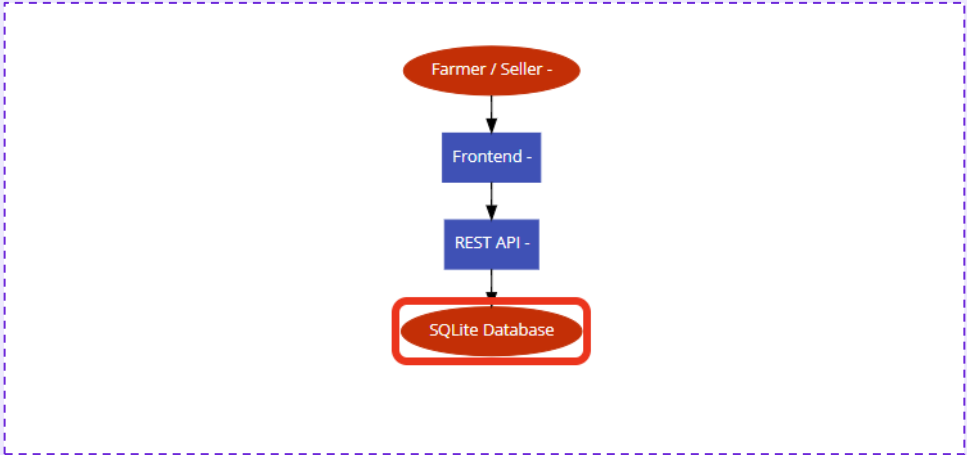
Future Scope

Enhancement 1: AI-based crop recommendations.
Enhancement 2: IoT-based soil monitoring.

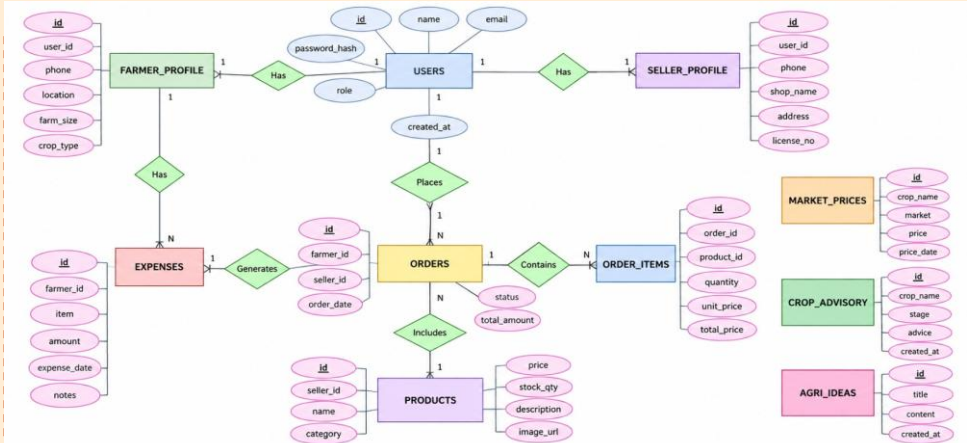
Tools & Technologies

Front-End: HTML, CSS, JavaScript
Back-End: Node.js + Express.js + REST APIs
Database: SQLite
Tools: Git & GitHub, VS Code, Postman

System Architecture



ER Diagram & Database Design



Implementation Screenshots

